

CSEC Mathematics
June 2013 – Paper 2
Solutions

SECTION I

Answer ALL questions in this section.

All working must be clearly shown.

1. (a) Using a calculator, or otherwise, calculate the EXACT value of

(i) $\frac{1\frac{4}{5} - \frac{1}{3}}{2\frac{2}{5}}$ [2]

$$\text{Numerator} = 1\frac{4}{5} - \frac{1}{3}$$

$$= \frac{9}{5} - \frac{1}{3}$$

$$= \frac{27-5}{15}$$

$$= \frac{22}{15}$$

$$\text{Denominator} = 2\frac{2}{5}$$

$$= \frac{12}{5}$$

Now,

$$\text{Numerator} \div \text{Denominator} = \frac{22}{15} \div \frac{12}{5}$$

$$= \frac{22}{15} \times \frac{5}{12}$$

$$= \frac{110}{180}$$

$$= \frac{11}{18}$$

$$\therefore \frac{1\frac{4}{5} - \frac{1}{3}}{2\frac{2}{5}} = \frac{11}{18}$$

(ii) $\sqrt{1.5625} + (0.32)^2$

Using a calculator,

$$\begin{aligned}\sqrt{1.5625} + (0.32)^2 &= 1.25 + 0.1024 \\ &= 1.3524\end{aligned}$$

(b) Smiley Orange Juice is sold in cartons of two different sizes at the prices shown in the table below.

Carton Size	Cost
350 ml	\$4.20
450 ml	\$5.13

Which size carton of orange juice is the BETTER buy?

Justify your answer.

[3]

If 350 ml is sold for \$4.20,

Then 1 ml will cost the buyer $\frac{\$4.20}{350} = \0.012 per ml

If 450 ml is sold for \$5.13,

Then 1 ml will cost the buyer $\frac{\$5.13}{450} = \0.0114 per ml

$\therefore$ The better buy is the 450 ml carton costing \$5.13.

(c) Faye borrowed \$9 600 at 8% per annum compound interest.

- (i) Calculate the interest on the loan for the first year. [1]

Interest for the first year = 8% of \$9600

$$\begin{aligned} &= \frac{8}{100} \times \$9600 \\ &= \$768 \end{aligned}$$

At the end of the first year, she repaid \$4 368.

- (ii) How much did she still owe at the beginning of the second year? [2]

At the end of the first year,

Amount owed = The sum borrowed + The interest acquired

$$\begin{aligned} &= \$9600 + \$768 \\ &= \$10\,368 \end{aligned}$$

At the beginning of the second year,

Amount owed = Total amount owed at the end of the first year – Amount repaid

$$\begin{aligned} &= \$10\,368 - \$4\,368 \\ &= \$6000 \end{aligned}$$

- (iii) Calculate the interest on the remaining balance for the second year.

[1]

Amount of interest on the remaining balance = 8% of \$6 000

$$= \frac{8}{100} \times \$6\,000$$

$$= \$480$$

Total: 11 marks

2. (a) Factorize completely:

(i) $2x^3 - 8x$ [2]

$$\begin{aligned} & 2x^3 - 8x \\ &= 2x(x^2 - 4) \\ &= 2x(x^2 - 2^2) \quad \text{[difference of two squares]} \\ &= 2x(x - 2)(x + 2) \end{aligned}$$

(ii) $3x^2 - 5x - 2$ [2]

$$\begin{aligned} & 3x^2 - 5x - 2 \\ &= 3x^2 - 6x + x - 2 \\ &= 3x(x - 2) + 1(x - 2) \\ &= (3x + 1)(x - 2) \end{aligned}$$

(b) (i) Make C the subject of the formula $F = \frac{9}{5}C + 32$. [2]

$$\begin{aligned} F &= \frac{9}{5}C + 32 \\ F - 32 &= \frac{9}{5}C \\ \frac{9}{5}C &= F - 32 \\ 9C &= 5(F - 32) \\ C &= \frac{5(F - 32)}{9} \end{aligned}$$

(ii) Given that $F = 113$, calculate the value of C .

[1]

When $F = 113$,

$$C = \frac{5(113-32)}{9}$$

$$C = \frac{5(81)}{9}$$

$$C = 45$$

(c) 500 tickets were sold for a concert. Of these x tickets were sold at \$6 each, and the remainder at \$10 each.

(i) Write an expression, in terms of x , for

(a) the number of tickets sold at \$10 each

[1]

The number of tickets sold = 500

The number of tickets sold at \$6 each = x

$\therefore$ The remaining tickets sold at \$10 each = $500 - x$

(b) the TOTAL amount of money collected for the sale of the 500 tickets.

[1]

The total cost of x tickets at \$6 each = $6x$

The cost of $(500 - x)$ tickets at \$10 each = $10(500 - x)$

$$\begin{aligned}
 \text{Thus, the total amount of money collected} &= 6x + 10(500 - x) \\
 &= 6x + 5000 - 10x \\
 &= 5000 - 4x
 \end{aligned}$$

(ii) The sum of \$4108 was collected for the sale of the 500 tickets.

Calculate the number of tickets sold at \$6 each.

[3]

$$5000 - 4x = 4108$$

$$4x = 5000 - 4108$$

$$4x = 892$$

$$x = \frac{892}{4}$$

$$x = 223$$

$\therefore$ 223 tickets were sold for \$6 each.

Total: 12 marks

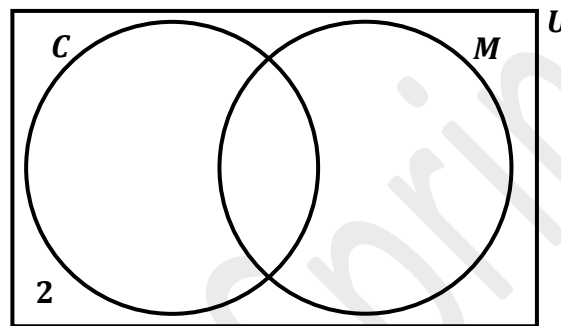
3. (a) A survey of the 30 students in Form 5 showed that some students used cameras (C) or mobile phones (M) to take photographs.

20 students used mobile phones

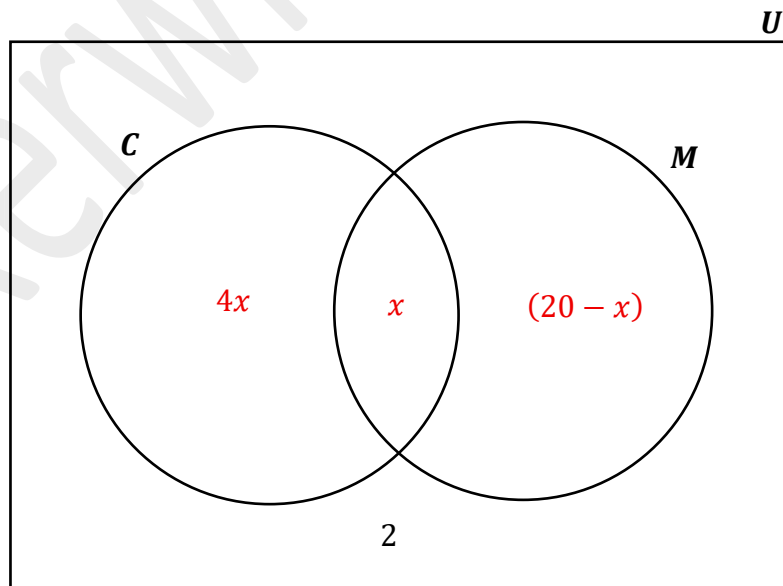
$4x$ students used ONLY cameras

x students used BOTH mobile phones and cameras

2 students did not use either cameras or phones.



- (i) Copy the Venn diagram below and complete it to show, in terms of x , the number of students in each region. [3]



- (ii) Write an expression, in terms of x , which represents the TOTAL number of students in the survey. [1]

$$\begin{aligned}\text{Total number of students} &= 2 + 4x + x + (20 - x) \\ &= 2 + 4x + x + 20 - x \\ &= 4x + 22\end{aligned}$$

- (iii) Determine the number of students in Form 5 who used ONLY cameras. [2]

The total number of students in the class is 30.

So,

$$4x + 22 = 30$$

$$4x = 30 - 22$$

$$4x = 8$$

$$x = \frac{8}{4}$$

$$x = 2$$

Therefore, the number of students who uses cameras only = $4x$

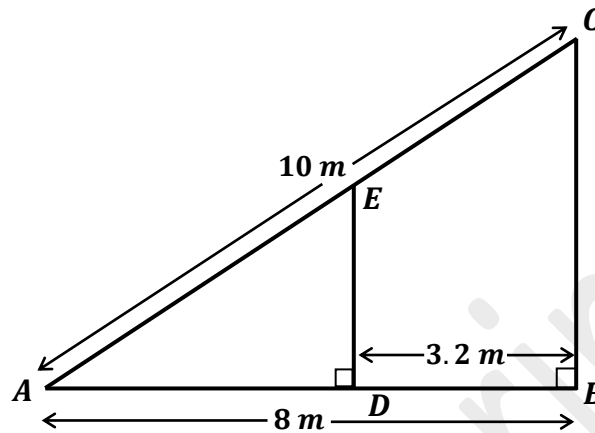
$$= 4(2)$$

$$= 8$$

(b) In the diagram below, **not drawn to scale**, AEC and ADB are straight lines.

$$\angle ABC = \angle ADE = 90^\circ$$

$$AC = 10\text{ m}, AB = 8\text{ m} \text{ and } DB = 3.2\text{ m}$$



(i) Calculate the length of BC .

[2]

Consider the triangle ABC .

Using Pythagoras' Theorem,

$$AB^2 + BC^2 = AC^2$$

$$BC^2 = AC^2 - AB^2$$

$$BC^2 = (10)^2 - (8)^2$$

$$BC^2 = 100 - 64$$

$$BC^2 = 36$$

$$BC = \sqrt{36}$$

$$BC = 6\text{ m}$$

- (ii) Explain why triangles ABC and ADE are similar.

[1]

The two triangles, ADE and ABC , have the same size of angles and are equiangular.

However, none of their sides are equal.

So, the two triangles are equiangular but not congruent.

Therefore, the two triangles ADE and ABC are similar since they possess the same shape but differ in size.

- (iii) Determine the length of DE .

[3]

Note that if two triangles are similar, then the ratio of their corresponding sides are equal.

$$AD = 8 - 3.2$$

$$= 4.8 \text{ m}$$

Hence,

$$\frac{AD}{AB} = \frac{DE}{BC}$$

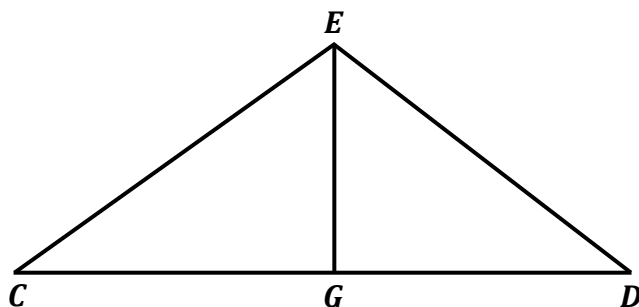
$$\frac{4.8}{8} = \frac{DE}{6}$$

$$DE = \frac{6 \times 4.8}{8}$$

$$DE = 3.6 \text{ m}$$

Total: 12 marks

4. (a) The diagram below shows an isosceles triangle CDE . G is the midpoint of CD .



- (i) Measure and state, in centimetres, the length of DE . [1]

By measurement with a ruler, $DE = 5 \text{ cm}$.

- (ii) Measure and state, in degrees, the size of $\angle ECD$. [1]

By measurement with a protractor, $\angle ECD = 37^\circ$.

- (iii) Determine the perimeter of the triangle CDE . [2]

By measurement with a ruler, $ED = EC = 5 \text{ cm}$ and $CD = 8 \text{ cm}$.

Therefore,

$$\begin{aligned} \text{The perimeter of triangle } CDE &= 5 + 5 + 8 \\ &= 18 \text{ cm} \end{aligned}$$

- (iv) Calculate the area of the triangle CDE .

[1]

By measurement with a ruler, $EG = 3 \text{ cm}$.

Since G is the midpoint of CD and $\triangle CDE$ is isosceles, then EG is perpendicular to CD .

Therefore,

$$\begin{aligned}\text{Area of } \triangle CDE &= \frac{b \times h}{2} \\ &= \frac{8 \times 3}{2} \\ &= \frac{24}{2} \\ &= 12 \text{ cm}^2\end{aligned}$$

- (b) $A(-1, 4)$ and $B(3, 2)$ are the end points of a line segment AB . Determine

- (i) the gradient of AB

[2]

$$\begin{aligned}\text{Gradient of } AB &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{2 - 4}{3 - (-1)} \\ &= \frac{-2}{4} \\ &= -\frac{1}{2}\end{aligned}$$

- (ii) the coordinates of the midpoint of AB

[2]

Let the midpoint of AB be M .

Using the midpoint formula, we get

$$\begin{aligned} M &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\ &= \left(\frac{-1 + 3}{2}, \frac{4 + 2}{2} \right) \\ &= \left(\frac{2}{2}, \frac{6}{2} \right) \\ &= (1, 3) \end{aligned}$$

- (iii) the equation of the perpendicular bisector of AB .

[3]

$$\begin{aligned} \text{The gradient of the perpendicular bisector} &= \frac{-1}{-\frac{1}{2}} \\ &= -1 \div -\frac{1}{2} \\ &= -1 \times -2 \\ &= 2 \end{aligned}$$

Thus, $m = 2$.

Substituting $m = 2$ and the midpoint $(1, 3)$ into $y = mx + c$, we get

$$3 = (2)(1) + c$$

$$3 = 2 + c$$

$$c = 3 - 2$$

$$c = 1$$

$\therefore$ Equation of the perpendicular bisector AB is $y = 2x + 1$.

Total: 12 marks

5. (a) The incomplete table below shows one pair of values for A and R where A is directly proportional to the square of R .

A	36		196
R	3	5	

- (i) Express A in terms of R and a constant k . [1]

$$A \propto R^2$$

$$\therefore A = kR^2 \text{ where } k \text{ is the constant of proportion}$$

- (ii) Calculate the value of the constant k . [2]

Substituting $A = 36$ and $R = 3$ gives,

$$36 = k(3)^2$$

$$36 = 9k$$

$$k = \frac{36}{9}$$

$$k = 4$$

(iii) Copy and complete the table.

$$A = 4R^2$$

When $R = 5$,

$$A = 4(5)^2$$

$$A = 4 \times 25$$

$$A = 100$$

When $A = 196$,

$$196 = 4R^2$$

$$R^2 = \frac{196}{4}$$

$$R^2 = 49$$

$$R = \sqrt{49}$$

$$R = 7$$

The completed table is shown below:

A	36	100	196
R	3	5	7

(b) Given that $f(x) = \frac{2x+1}{3}$ and $g(x) = 4x + 5$, determine the values of:

(i) $fg(2)$ [3]

$$g(2) = 4(2) + 5$$

$$= 8 + 5$$

$$= 13$$

$$\therefore fg(2) = f(13)$$

$$= \frac{2(13)+1}{3}$$

$$= \frac{26+1}{3}$$

$$= \frac{27}{3}$$

$$= 9$$

(ii) $f^{-1}(3)$ [3]

$$f(x) = \frac{2x+1}{3}$$

$$\text{Let } y = f(x).$$

$$y = \frac{2x+1}{3}$$

Interchange the variables x and y :

$$x = \frac{2y+1}{3}$$

Make y the subject:

$$3x = 2y + 1$$

$$2y = 3x - 1$$

$$y = \frac{3x-1}{2}$$

$$\therefore f^{-1}(x) = \frac{3x-1}{2}$$

Now,

$$f^{-1}(3) = \frac{3(3)-1}{2}$$

$$= \frac{9-1}{2}$$

$$= \frac{8}{2}$$

$$= 4$$

Total: 11 marks

6. (a) A car, travelling along a straight road at a constant speed of 54 km/h , takes 20 seconds to travel the distance between two sign posts.

Calculate

- (i) the speed of the car in m/s [2]

Recall:

$$1\text{km} = 1000\text{m}$$

$$1 \text{ hour} = 3600 \text{ seconds}$$

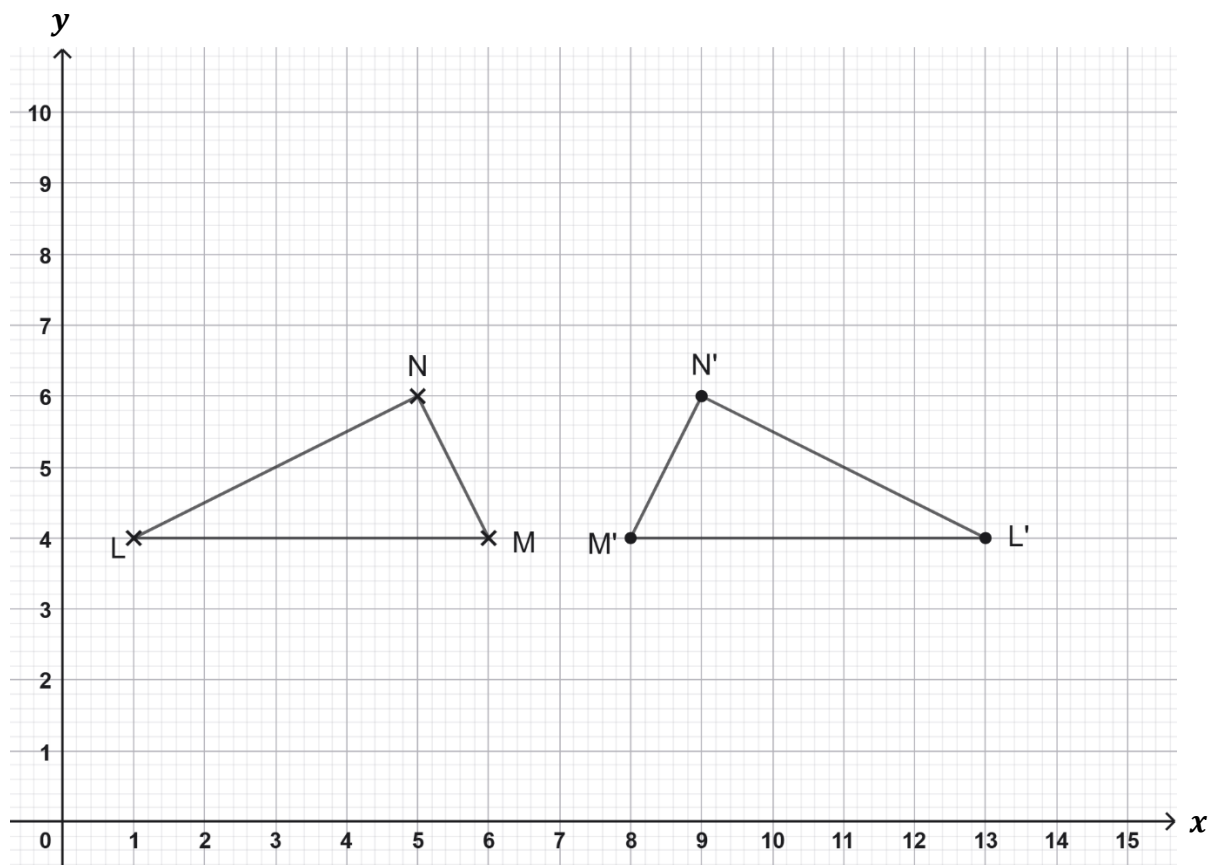
$$\begin{aligned} 54 \text{ kmh}^{-1} &= 54 \times \frac{1000}{3600} \\ &= 15 \text{ ms}^{-1} \end{aligned}$$

- (ii) the distance, in metres, between the two sign posts [2]

$$\begin{aligned} \text{Distance} &= \text{Speed} \times \text{Time} \\ &= 15 \times 20 \\ &= 300 \text{ m} \end{aligned}$$

(b) An answer sheet is provided for this question.

The graph below shows triangle LMN and its image $L'M'N'$ after undergoing a single transformation.

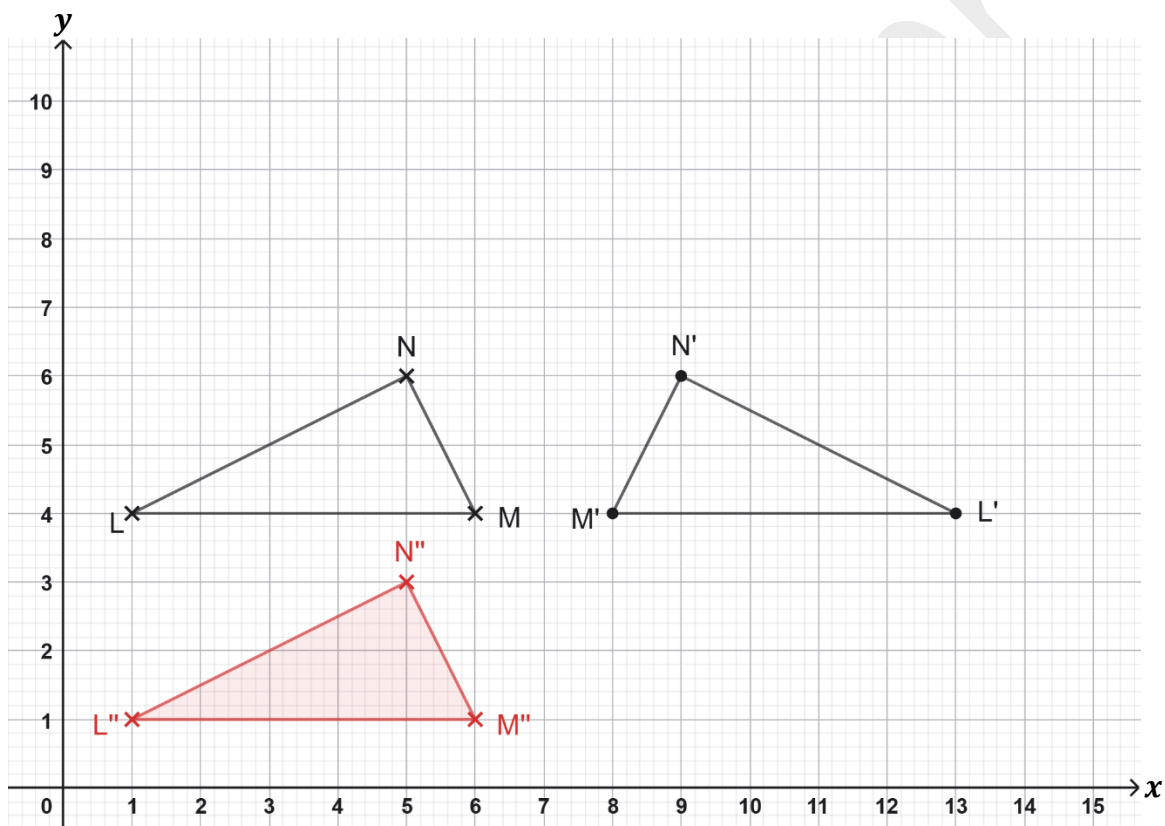


(i) Describe **fully** the transformation that maps $\triangle LMN$ onto $\triangle L'M'N'$. [2]

The image and the object are congruent. Also, the image is laterally inverted. Therefore, the transformation is a reflection in the line $x = 7$.

- (ii) On the answer sheet provided, draw triangle $L''M''N''$, the image of triangle LMN , after a translation by the vector $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$. [2]

Triangle $L''M''N''$ is shown below:



- (iii) Name and describe a combination of TWO transformations which may be used to map $\Delta L''M''N''$ onto $\Delta L'M'N'$. [3]

Two transformations which may be used to map $\Delta L''M''N''$ onto $\Delta L'M'N'$:

1) Translate $L''M''N''$ by the vector $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$ then

2) Reflect LMN in the line $x = 7$

Total: 11 marks

7. The table below shows the amount, to the nearest dollar, spent by a group of 40 students at the school canteen during a period of one week.

Amount Spent (\$)	Number of Students	Cumulative Frequency
1 – 10	3	3
11 – 20	7	10
21 – 30	9	19
31 – 40	11	
41 – 50	8	
51 – 60	2	

(a) Copy and complete the table to show the cumulative frequency.

[2]

The completed table is shown below:

Amount Spent (\$)	Number of Students	Cumulative Frequency
1 – 10	3	3
11 – 20	7	10
21 – 30	9	19
31 – 40	11	30
41 – 50	8	38
51 – 60	2	40

(b) Using a scale of 1 *cm* to represent \$5 on the horizontal axis and 1 *cm* to

represent 5 students on the vertical axis, draw the cumulative frequency graph

for the data.

[5]

(Marks will be awarded for axes appropriately labelled, points correctly plotted, and a smooth curve carefully drawn.)

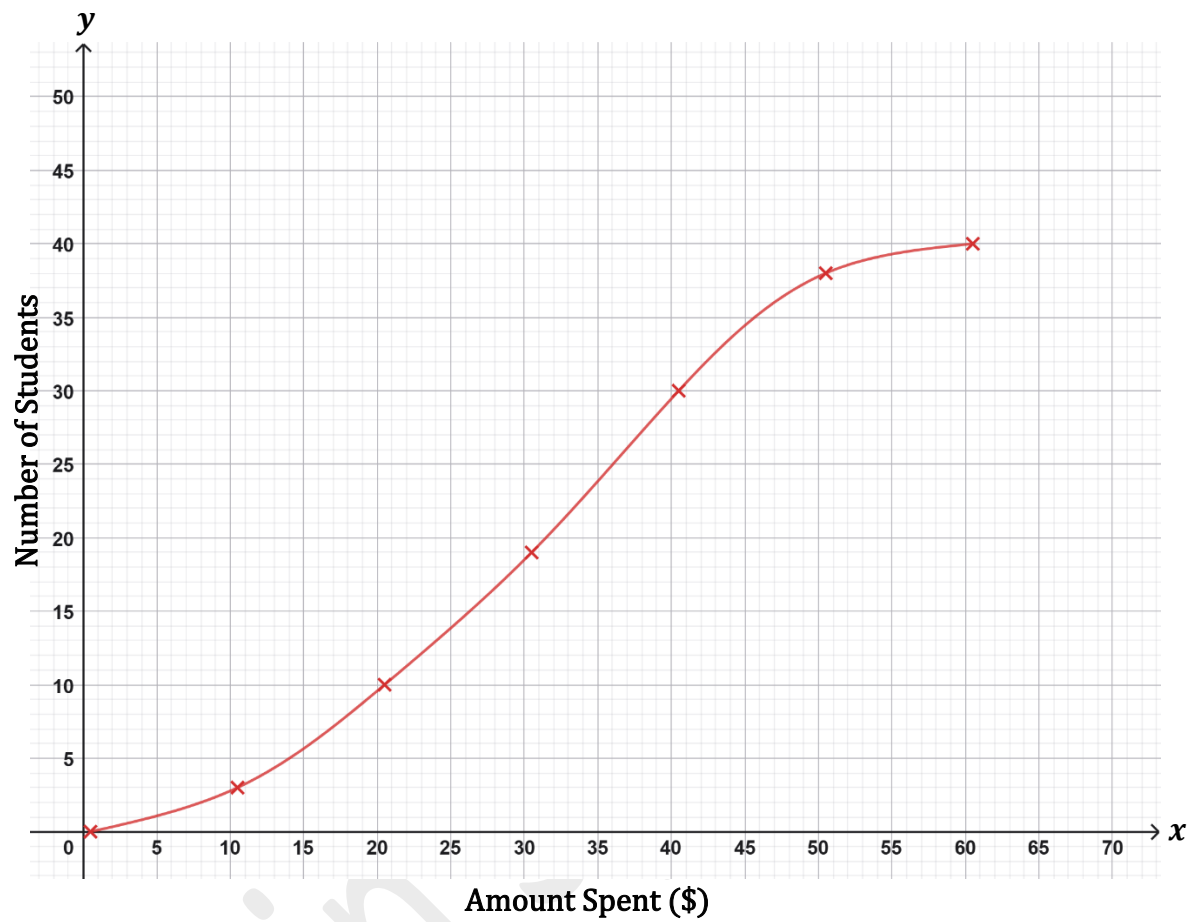
Amount Spent (x) L.C.L–U.C.L	Lower Class Boundary & Upper Class Boundary	Number of Students Frequency (f)	Cumulative Frequency	Points to be Plotted
				(0.5, 0)
1-10	$0.5 \leq x < 10.5$	3	3	(10.5, 3)
11-20	$10.5 \leq x < 20.5$	7	10	(20.5, 10)
21-30	$20.5 \leq x < 30.5$	9	19	(30.5, 19)
31-40	$30.5 \leq x < 40.5$	11	30	(40.5, 30)
41-50	$40.5 \leq x < 50.5$	8	38	(50.5, 38)
51-60	$50.5 \leq x < 60.5$	2	40	(60.5, 40)

Scale Used:

x -axis: 1 *cm* represents \$5

y -axis: 1 *cm* represents 5 students

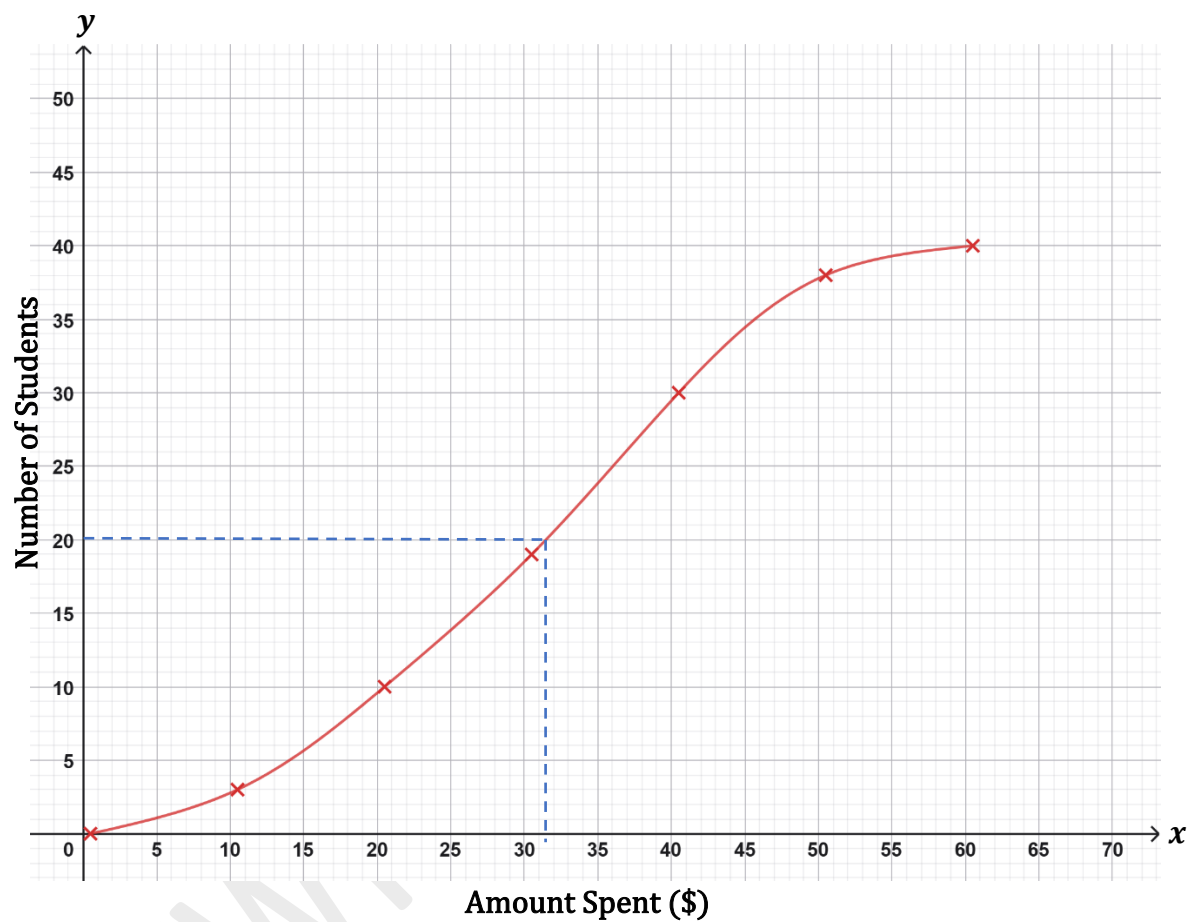
The cumulative frequency graph is shown on the next page:



(c) Use your graph to estimate

(i) the median amount of money spent

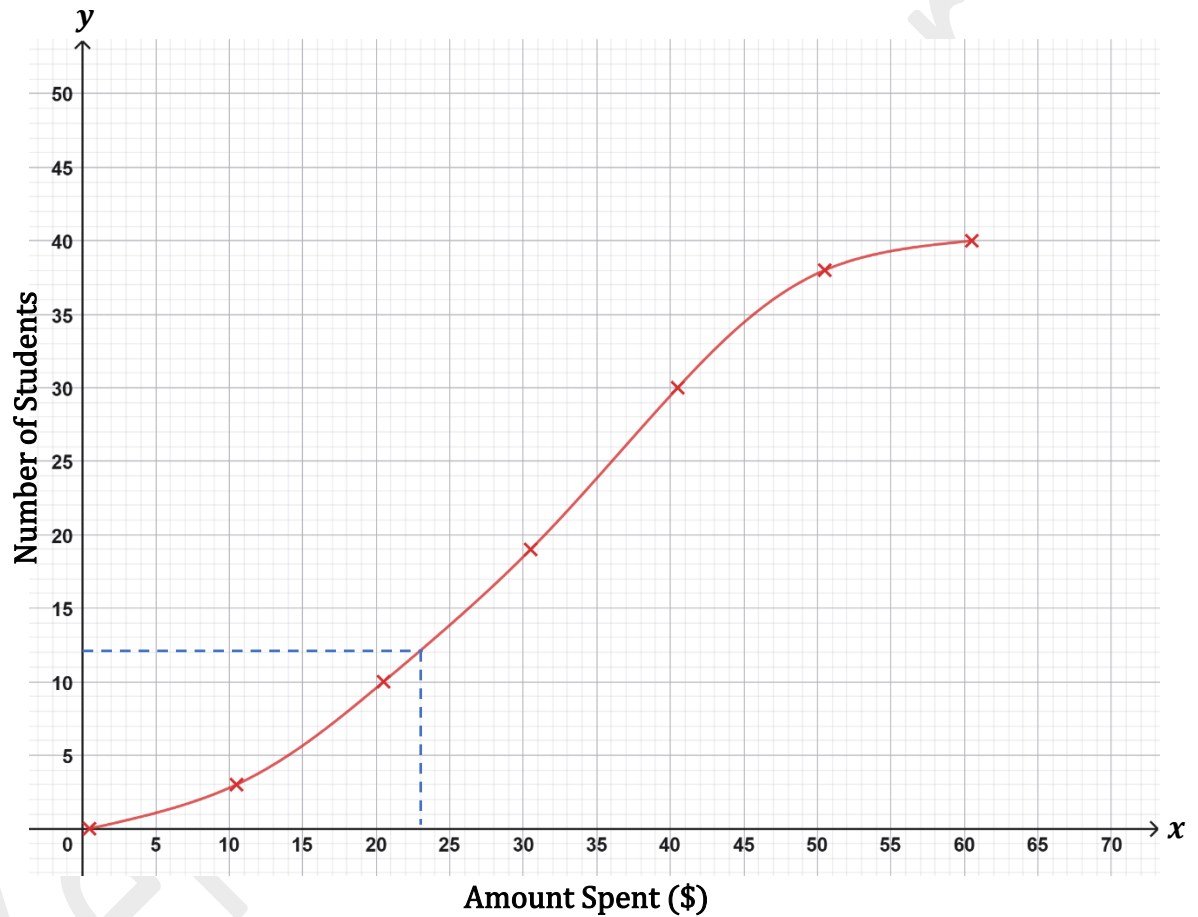
[2]



The median amount of money spent = \$31.50

- (ii) the probability that a student chosen at random spent less than \$23 during the week. [2]

Show on your graph, using broken lines, how these estimates were determined.



$$\begin{aligned}
 P(\text{student spends less than \$23}) &= \frac{\text{Number of students who spends less than \$23}}{\text{Total number of students}} \\
 &= \frac{12}{40} \\
 &= \frac{3}{10}
 \end{aligned}$$

Total: 11 marks

8. An answer sheet is provided for this question.

The drawings below show the first three diagrams in a sequence.

Diagram 1

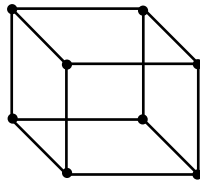


Diagram 2

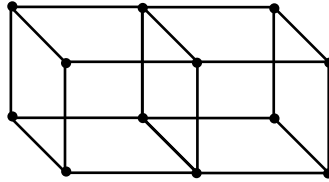
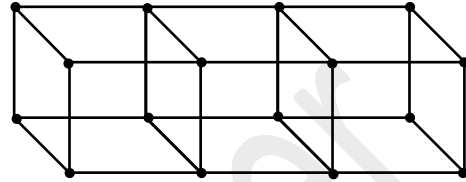
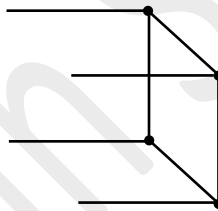


Diagram 3



Each diagram is made up of wires of equal length which are joined at the ends by balls of plasticine. Diagram 1 is made of 12 wires and 8 balls. Each new diagram in the sequence is formed by fitting the frame shown below to the right of the previous diagram.



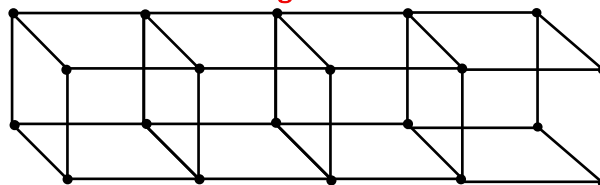
Thus, Diagram 2 has 8 more wires and 4 more balls than Diagram 1.

On the answer sheet provided:

(a) Draw a sketch of Diagram 4, the fourth diagram in the sequence.

[2]

Diagram 4



(b) Complete the table by inserting the missing values at the rows marked

(i) and (ii).

	Name of Diagram (N)	No. of Wires (W)	No. of Balls (B)	
	1	12	8	
	2	20	12	
	3	28	16	
(i)	4	36	20	[2]
(ii)	20	164	84	[4]

For the N th diagram,

$$W = 8N + 4 \quad \text{and} \quad B = 4N + 4$$

When $N = 4$,

$$W = 8(4) + 4 \quad \text{and} \quad B = 4(4) + 4$$

$$W = 32 + 4 \quad B = 16 + 4$$

$$W = 36 \quad B = 20$$

When $N = 20$,

$$W = 8(20) + 4 \quad \text{and} \quad B = 4(20) + 4$$

$$W = 160 + 4 \quad B = 80 + 4$$

$$W = 164 \quad B = 84$$

(c) Write the rules which may be used to find the values of W and of B where N is known.

(i) $W =$ _____ [1]

The rule to find W is: $W = 8N + 4$

(ii) $B =$ _____ [1]

The rule to find B is: $B = 4N + 4$

Total: 10 marks

SECTION II

Answer TWO questions in this section.

ALGEBRA AND RELATIONS, FUNCTIONS AND GRAPHS

9. (a) An answer sheet is provided for this question.

Trish wishes to buy x oranges and y mangoes which she intends to carry in her bag. Her bag has space for only 6 fruits.

- (i) Write an inequality to represent this information. [1]

Number of oranges = x

Number of mangoes = y

Total number of fruits = $x + y$

Since the total number of oranges and mangoes must not exceed 6, then the inequality to represent this information is:

$$x + y \leq 6$$

To get a good bargain, she must buy AT LEAST 2 mangoes.

- (ii) Write an inequality to represent this information. [1]

Since Trish must buy at least 2 mangoes, then the inequality to represent this information is:

$$y \geq 2$$

More information about the number of oranges and mangoes associated with the good bargain is represented by

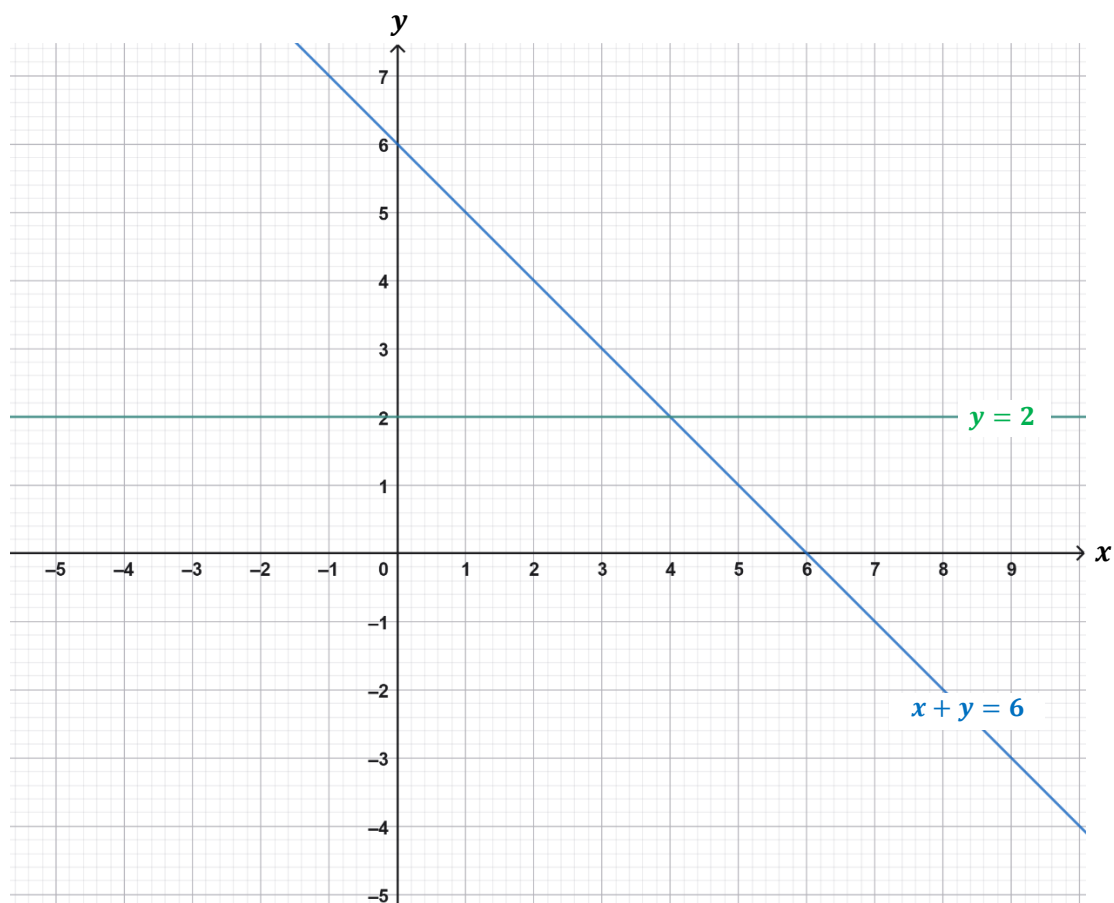
$$y \leq 2x.$$

- (iii) Write the information represented by this inequality as a sentence in your own words. [2]

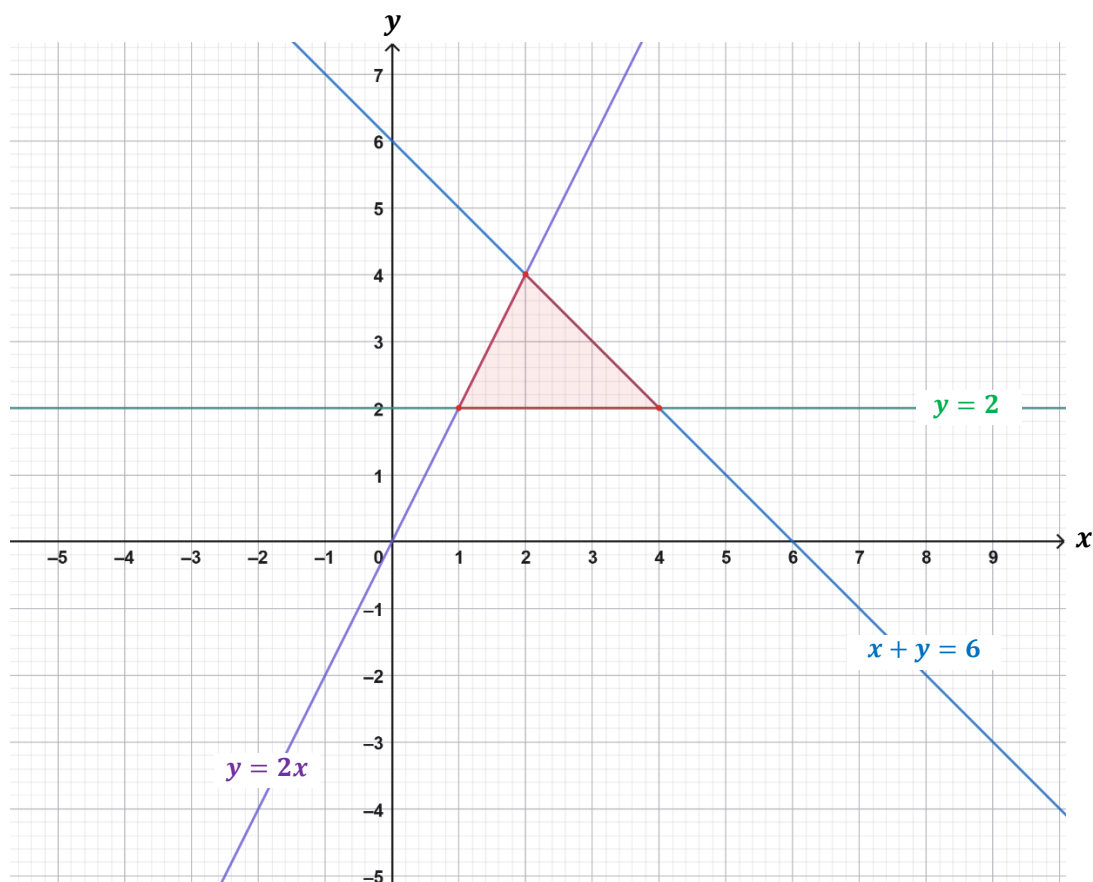
Inequality: $y \leq 2x$

In words, the number of mangoes is less than or equal to two times the number of oranges.

- (iv) On the answer sheet provided, draw the lines associated with the two inequalities obtained in (i) and (ii) above. [3]



- (v) Shade on your graph the region which represents the solution set for the three inequalities. [1]



(b)(i) Write $3x^2 - 12x + 8$ in the form $a(x + h)^2 + k$, where a , h and k are constants. [3]

$$\begin{aligned}
 3x^2 - 12x + 8 &= 3(x^2 - 4x) + 8 \\
 &= 3(x^2 - 4x + 4) + 8 - 3(4) \\
 &= 3(x - 2)^2 + 8 - 12 \\
 &= 3(x - 2)^2 - 4 \quad \text{which is of the form } a(x + h)^2 + k \\
 &\quad \text{where } a = 3, h = -2 \text{ and } k = -4
 \end{aligned}$$

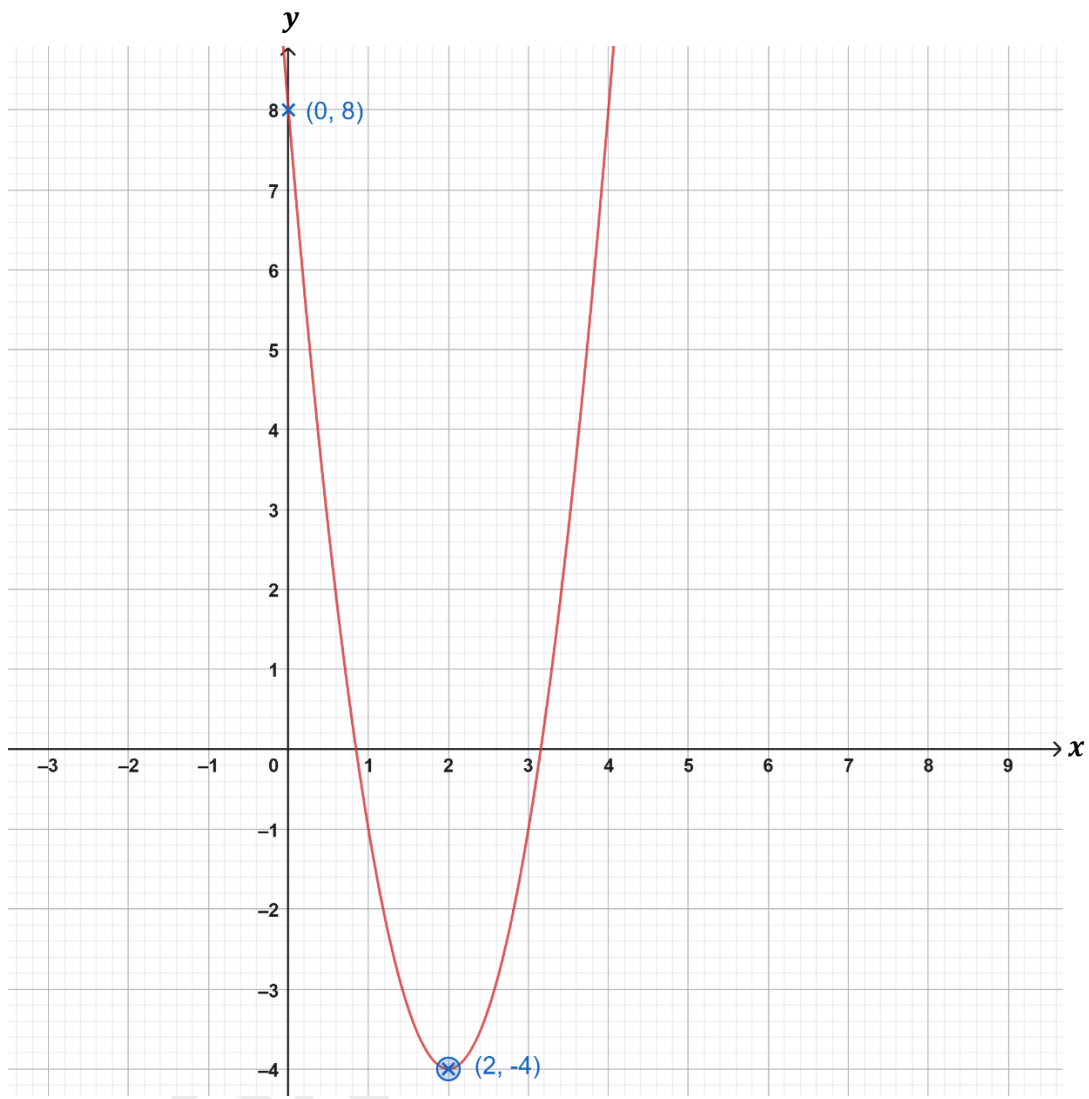
(ii) Sketch the graph of $y = 3x^2 - 12x + 8$, showing on your sketch

(a) the intercept on the y -axis

(b) the coordinates of the minimum point. [4]

The graph of $y = 3x^2 - 12x + 8$ is shown on the next page.

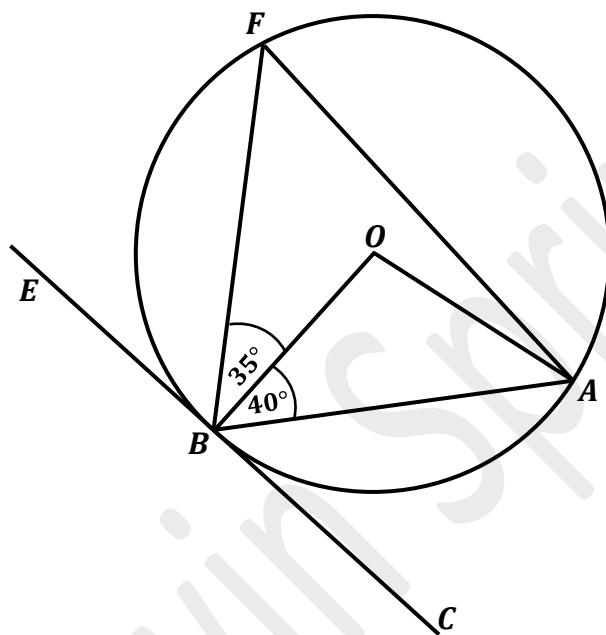
From the graph, the y -intercept is $(0, 8)$ and the coordinates of the minimum point is $(2, -4)$



Total: 15 marks

MEASUREMENT, GEOMETRY AND TRIGONOMETRY

10. (a) The diagram below, **not drawn to scale**, shows a circle with centre O . EBC is a tangent to the circle. $\angle OBA = 40^\circ$ and $\angle OBF = 35^\circ$.



Calculate, **giving reasons for your answer**, the measure of

- (i) $\angle EBF$

[1]

The angle made by a tangent to a circle and any chord, at the point of contact, is 90° .

So, $\angle OBE = 90^\circ$.

OB is the radius and EBC is the tangent. The point of contact is B .

$$\therefore \angle EBF = 90^\circ - 35^\circ$$

$$= 55^\circ$$

(ii) $\angle BOA$

[2]

Since OB and OA are both radii of the same circle, then $OB = OA$.

The base angles of an isosceles triangle are equal.

$\therefore \triangle OAB$ is an isosceles triangle with $\angle OBA = \angle OAB = 40^\circ$.

The sum of angles in a triangle add up to 180° .

$$\therefore \angle BOA = 180^\circ - (40^\circ + 40^\circ)$$

$$= 180^\circ - 80^\circ$$

$$= 100^\circ$$

(iii) $\angle AFB$

[2]

The angle subtended by a chord at the center of a circle is twice the angle that the chord subtends at the circumference, standing on the same arc.

So,

$$\angle AFB = \frac{1}{2} (100^\circ)$$

$$= 50^\circ$$

(iv) $\angle OAF$

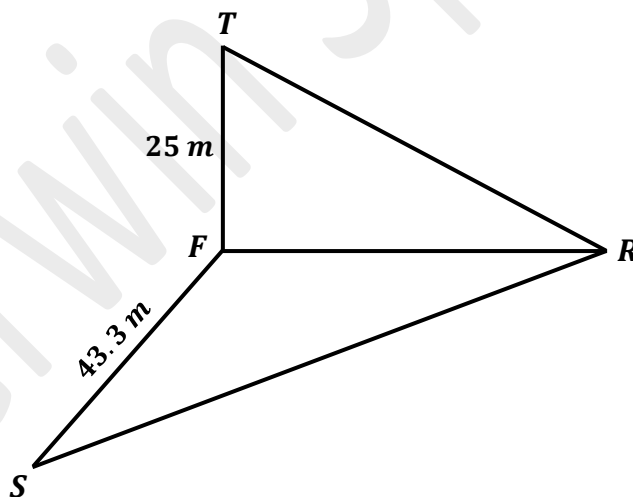
The sum of angles in a triangle add up to 180° .

$$\therefore \angle OAF = 180^\circ - (50^\circ + 35^\circ + 40^\circ + 40^\circ)$$

$$= 180^\circ - 165^\circ$$

$$= 15^\circ$$

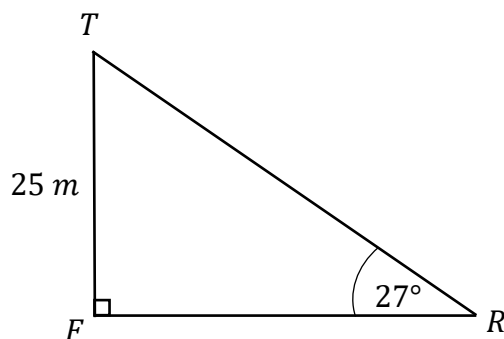
- (b) The diagram below, **not drawn to scale**, shows three points R, S and F on the horizontal ground. FT is a vertical tower of height 25 m. The angle of elevation of the top of the tower, T , from R is 27° . R is due east of F and S is due south of F . $SF = 43.3$ m.



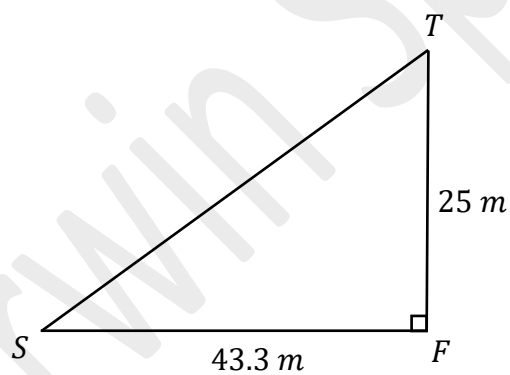
- (i) Sketch **separate** diagrams of the triangles RFT , TFS and SFR . Mark on EACH diagram the given measures of sides and angles. [3]

The diagrams are as follows.

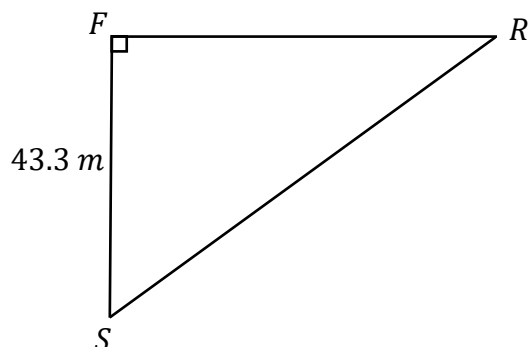
For triangle RFT :



For triangle TFS :

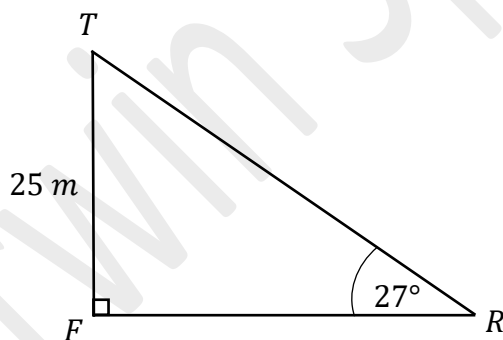


For triangle SFR :



- (ii) Show, by calculation, that $RF = 49.1$ m. [2]

Consider triangle RFT :



$$\tan 27^\circ = \frac{25}{RF}$$

$$RF = \frac{25}{\tan 27^\circ}$$

$$RF = 49.1 \text{ m}$$

- (iii) Calculate the length of SR correct to 1 decimal place.

Consider the triangle SFR .

Using Pythagoras' Theorem,

$$SR^2 = FR^2 + FS^2$$

$$= (49.06)^2 + (43.3)^2$$

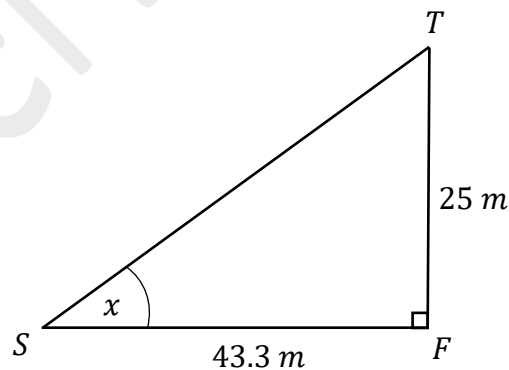
$$= 4281.77$$

$$SR = \sqrt{4281.77}$$

$$= 65.4 \text{ m} \quad (\text{to 1 decimal place})$$

- (iv) Calculate the angle of elevation of the top of the tower, T , from S . [2]

Consider triangle TFS :



$$\tan x = \frac{25}{43.3}$$

$$x = \tan^{-1} \left(\frac{25}{43.3} \right)$$

$$x = 30.0^\circ \quad (\text{to 1 decimal place})$$

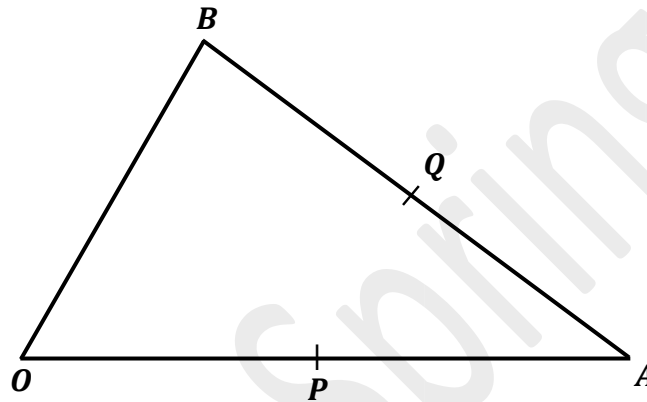
Total: 15 marks

Kerwin Springer

VECTORS AND MATRICES

11. (a) In the diagram below, **not drawn to scale**, P and Q are the midpoints of OA and AB respectively.

$$\overrightarrow{OA} = 2\mathbf{a} \quad \text{and} \quad \overrightarrow{OB} = 2\mathbf{b}.$$



- (i) Express in terms of $\mathbf{a}$ and $\mathbf{b}$ the vectors

(a) $\overrightarrow{AB}$

[2]

Using the vector triangle law, we get

$$\begin{aligned} \overrightarrow{AB} &= \overrightarrow{AO} + \overrightarrow{OB} \\ &= -(2\mathbf{a}) + 2\mathbf{b} \\ &= -2\mathbf{a} + 2\mathbf{b} \end{aligned}$$

(b) $\overrightarrow{PQ}$

$$\overrightarrow{OP} = \overrightarrow{PA} = \mathbf{a}$$

$$\begin{aligned}\overrightarrow{AQ} &= \frac{1}{2}\overrightarrow{AB} \\ &= \frac{1}{2}(-2\mathbf{a} + 2\mathbf{b})\end{aligned}$$

$$\begin{aligned}\overrightarrow{PQ} &= \overrightarrow{PA} + \overrightarrow{AQ} \\ &= \mathbf{a} + \frac{1}{2}(-2\mathbf{a} + 2\mathbf{b}) \\ &= \mathbf{a} - \mathbf{a} + \mathbf{b} \\ &= \mathbf{b}\end{aligned}$$

(ii) State TWO geometrical relationships that exist between OB and PQ .

Give reasons for your answers.

[2]

$$OB = 2PQ$$

Since PQ can be represented as a scalar multiple of AB , then OB and PQ are parallel.

Therefore, $|OB| = 2|PQ|$.

That is, OB is twice the length of PQ .

(b) Given that $M = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}$

(i) Evaluate M^{-1} , the inverse of M .

[2]

$$\det M = ad - bc$$

$$= (2)(3) - (1)(4)$$

$$= 6 - 4$$

$$= 2$$

$$\text{adj}(M) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$= \begin{pmatrix} 3 & -1 \\ -4 & 2 \end{pmatrix}$$

$$\therefore M^{-1} = \frac{1}{\det M} \times \text{adj}(M)$$

$$= \frac{1}{2} \begin{pmatrix} 3 & -1 \\ -4 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{3}{2} & -\frac{1}{2} \\ -2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{3}{2} & -\frac{1}{2} \\ -2 & 1 \end{pmatrix}$$

(ii) Show that $M^{-1}M = I$.

$$\begin{aligned}
 M^{-1}M &= \begin{pmatrix} \frac{3}{2} & -\frac{1}{2} \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} \\
 &= \frac{1}{2} \begin{pmatrix} 3 & -1 \\ -4 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} \\
 &= \frac{1}{2} \begin{pmatrix} (3 \times 2) + (-1 \times 4) & (3 \times 1) + (-1 \times 3) \\ (-4 \times 2) + (2 \times 4) & (-4 \times 1) + (2 \times 3) \end{pmatrix} \\
 &= \frac{1}{2} \begin{pmatrix} 6 - 4 & 3 - 3 \\ -8 + 8 & -4 + 6 \end{pmatrix} \\
 &= \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\
 &= I
 \end{aligned}$$

$$\therefore M^{-1}M = I$$

(iii) Use a matrix method to solve for r, s, t and u in the equation

$$\begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} r & s \\ t & u \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix}. \quad [5]$$

$$M^{-1} \times M \times \begin{pmatrix} r & s \\ t & u \end{pmatrix} = M^{-1} \times \begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix}$$

Therefore,

$$\begin{pmatrix} r & s \\ t & u \end{pmatrix} = M^{-1} \times \begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix}$$

$$\begin{aligned}
 &= \frac{1}{2} \begin{pmatrix} 3 & -1 \\ -4 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix} \\
 &= \frac{1}{2} \begin{pmatrix} (3 \times 2) + (-1 \times 4) & (3 \times 1) + (-1 \times -1) \\ (-4 \times 2) + (2 \times 4) & (-4 \times 1) + (2 \times -1) \end{pmatrix} \\
 &= \frac{1}{2} \begin{pmatrix} 6 - 4 & 3 + 1 \\ -8 + 8 & -4 - 2 \end{pmatrix} \\
 &= \frac{1}{2} \begin{pmatrix} 2 & 4 \\ 0 & -6 \end{pmatrix} \\
 &= \begin{pmatrix} \frac{2}{2} & \frac{4}{2} \\ \frac{0}{2} & \frac{-6}{2} \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 2 \\ 0 & -3 \end{pmatrix}
 \end{aligned}$$

Equating the corresponding entries, we get:

$$r = 1$$

$$s = 2$$

$$t = 0$$

$$u = -3$$

Total: 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.